DATA 442/642 · Advanced Machine Learning · Week 2

2.2 Unconstrained Methods

Line search, steepest descent, momentum, and SGD.

1. Line search

Almost every method in this note is

$$ \boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} + \alpha^{(k)}\mathbf{s}^{(k)}. $$

1. Pick a **direction** $\mathbf{s}^{(k)}$.
2. Pick a **step** $\alpha^{(k)}$ (learning rate) so that J decreases along that ray.
3. Take the step.

A direction is a **descent** direction when $\mathbf{s}^\top\nabla J < 0$. Steepest descent uses $\mathbf{s}=-\nabla J$.

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2. Why steepest descent zigzags

On a quadratic $J(\boldsymbol{\theta})=\tfrac12\boldsymbol{\theta}^\top\mathbf{Q}\boldsymbol{\theta}$ with $\mathbf{Q}$ SPD, the contours are ellipses. The axis ratio is the **condition number** of $\mathbf{Q}$ (largest over smallest eigenvalue). A long thin valley makes $-\nabla J$ point into the walls. Exact line search still hops. That is not a bug in your code. It is the geometry.

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3. Momentum

Remember the last step and refuse to turn on a dime:

$$ \boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} -\alpha^{(k)}\nabla J(\boldsymbol{\theta}^{(k)}) +\beta\Delta\boldsymbol{\theta}^{(k)}, $$

with $\beta\in[0,1]$. A heavy ball in a valley: oscillations damp, progress along the floor improves. Noisy gradients (SGD) benefit because the memory averages them.

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4. SGD and mini-batches

If $J=\sum_i j_i$, **batch** GD uses ∇J (all i). **SGD** uses one j_i (or a random subset). **Mini-batch** sits in between (Lab 2 uses size 20).

SGD is cheaper per step and noisier. The noise is sometimes useful: it keeps you from sitting in a sharp hole. The path in (\boldsymbol{\theta})-space looks like a drunk walk toward the bowl, not a straight slide.

Lab 2 asks you to draw those three paths on linear-looking data and to say which one arrives.